# Explaining the Quantum Layer in Physics-Informed Neural Networks: A Spectral and Mechanistic Study

WISER Global Quantum+AI Program 2026 — BQP Challenge
*Quantum-Assisted Physics-Informed Neural Networks for CFD*

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**Abstract**

Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) replace the first hidden layer of a PINN with a variational quantum circuit (VQC). Prior work reports that this maintains accuracy while reducing trainable parameters, but the *mechanism* — when, why, and how the quantum layer changes learning — is unexplained. We build a verified, parameter-matched, PDE-agnostic test bench (viscous Burgers and heat, each with an exact reference) and a five-dimension explainability (XAI) suite, and use them to ask a single sharp question: is the quantum layer's behaviour governed by the *frequency band* it adds (the Schuld–Sweke–Meyer Fourier picture), or by something else? An initial one-knob-at-a-time ablation suggested a "bandwidth-gated advantage", but a subsequent *adversarial self-audit* found those runs were undertrained (L-BFGS disabled; every Burgers model stuck at rel-$L_2 \approx 0.47$, the shock unresolved by any model), making the comparisons differences between failing models. We therefore re-ran converged cores (L-BFGS on) at two model scales, with multi-seed statistics and fair controls. The result is a consistent, honest negative. **(1) At small (121-param) scale**, the classical PINN (rel-$L_2$ $0.248 \pm 0.031$) beats every QAPINN. **(2) At resolved (1273-param) scale**, the QAPINN appears to win — but only against a *shallow* classical net: QAPINN-none reaches $0.094 \pm 0.075$ vs a shallow classical's 0.24 ($p = 0.003$), yet a *depth-matched* classical net ties it ($0.133 \pm 0.047$, $p = 0.35$), showing the gain is architectural (depth, post-net width) and classically reproducible, *not* quantum-specific. **(3) No robust entanglement effect:** an $n = 3$ "entanglement hurts" signal ($p \approx 0.02$) *vanished* at $n = 6$ ($p = 0.41$) — a lucky draw we retract; at the larger scale the entanglement gap is likewise non-significant ($p = 0.44$). We also correct a framing error: the verified "band = depth" fact governs the *VQC features*, not the full model (the tanh post-net reaches $k = 22$ once trained). The pipeline is independently verified (8/8) and every number matches its log. The contribution is a rigorous negative result and the controlled methodology — matched, structural, and depth-matched controls, multi-seed statistics, and an adversarial audit — that repeatedly exposed over-claims the QAPINN literature usually leaves standing.

## Contents

**1 Introduction** **3**

**2 Background and Related Work** **3**

**3 Methodology** **4**
3.1 Test problems . . . . . 4

3.2 Models . . . . . 4
3.3 Explainability suite . . . . . 4
3.4 Pipeline verification . . . . . 5

**4 Experiments and Results** **5**
4.1 Baseline comparison . . . . . 5
4.2 Tier-1 ablation . . . . . 5
4.3 Tier-3 ablation: qubits, measurement, encoding scale . . . . . 6
4.4 How the quantum layer changes the classical network . . . . . 7

**5 Converged Re-analysis (Post-Audit)** **8**

**6 Discussion** **11**

**7 Limitations** **12**

**8 Conclusion** **12**

**A Reproducibility** **13**

# 1 Introduction

Physics-Informed Neural Networks (PINNs) [1] solve ODEs/PDEs by making the differential equation itself the loss: a fully-connected network $u_\theta(x,t)$ is trained to minimise the PDE residual together with initial and boundary conditions. A Quantum-Assisted PINN (QAPINN) [4] is the hybrid variant obtained by *replacing the first hidden layer with a VQC*. The reference benchmark reports up to $\sim$20% fewer trainable parameters at maintained accuracy on 1-D Burgers.

The BQP challenge asks not to prove QAPINNs beat PINNs, but to *explain* the learning dynamics and to derive a methodology for problem-specific circuit design. This report takes the dominant theoretical lens for VQCs — that a data-encoding circuit is a truncated Fourier series [2] — and asks whether it is sufficient to explain a QAPINN's behaviour, and how the quantum layer changes the classical network it is attached to.

**Contributions.**

1. A verified, parameter-matched, PDE-agnostic bench (Burgers + heat, exact references) and a five-dimension XAI suite (spectral / F-principle, neurons, VQC internals, loss landscape, generalisation).
2. An empirical confirmation, on our own implementation, that the VQC is a truncated Fourier series whose bandwidth equals the re-upload depth.
3. A Tier-1 ablation (depth, entanglement, post-net width) isolating whether the advantage is a bandwidth effect or an entanglement effect.
4. A classical-side analysis quantifying how the quantum layer changes the downstream classical network's neurons, weights, and gradient dynamics.

# 2 Background and Related Work

**VQCs as Fourier series.** Schuld, Sweke and Meyer [2] prove that a VQC with data-encoding gates is a partial Fourier series $f(x) = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}$, whose accessible frequency set $\Omega$ is fixed by the eigenvalues of the encoding generators, while the coefficients $c_\omega$ are set by the trainable gates. Repeating (re-uploading) the encoding enlarges $\Omega$; with $L$ re-uploads of a Pauli-$Y$ rotation the accessible frequencies are the integers $\{-L,\dots,L\}$. This is the analysis language of the present work.

**Frequencies are not enough.** Mhiri et al. [5] refine this: the achievable *coefficients* are constrained (a coefficient's variance scales with its frequency's redundancy) and can vanish exponentially with qubit count. Having a frequency available is necessary but not sufficient to realise it.

**Quantum solvers and QAPINNs.** Kyriienko et al. [3] solve nonlinear DEs with differentiable quantum circuits using analytic derivatives. Hybrid QAPINN/QCPINN designs [4, 6] report parameter reduction at matched-or-better accuracy, but gains are PDE-dependent (quantum sometimes loses, e.g. on wave and cavity flow [6]), single-group, and rarely compute-matched or reproduced. A distinct trainability pathology — a "black-hole" collapse to the trivial solution — has recently been named [7]. Across the field, quantum gains are attributed to "entanglement-driven expressibility" as a heuristic, with no validated mechanism — the gap this study targets.

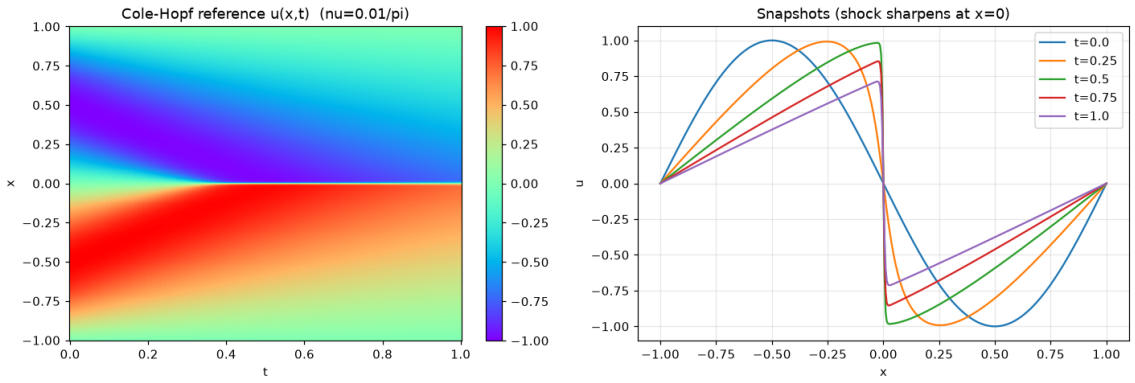



Figure 1: Exact Cole–Hopf reference for viscous Burgers. Left: field $u(x,t)$; right: snapshots showing the shock forming at $x = 0$. This is the self-generated ground truth used for all relative-$L_2$ scoring.

# 3 Methodology

## 3.1 Test problems

Both PDEs share the same domain $x \in [-1, 1]$, $t \in [0, 1]$, initial condition $u(x, 0) = -\sin(\pi x)$, and homogeneous Dirichlet BCs, so the *only* difference is the operator — hence the spectral content of the solution:

$$\text{Burgers:} \quad u_t + u\,u_x = \nu\,u_{xx}, \quad \nu = 0.01/\pi \qquad \text{(shock} \to \text{broadband)} \tag{1}$$

$$\text{Heat:} \quad u_t = \alpha\,u_{xx}, \quad \alpha = 1/\pi^2 \qquad \text{(single mode } k{=}1,\ u = -\sin(\pi x)e^{-t}\text{)} \tag{2}$$

Burgers uses an exact Cole–Hopf reference (Gauss–Hermite quadrature, machine precision); heat is analytic. Figure 1 shows the Burgers reference: a sine wave steepening into a shock at $x = 0$.

## 3.2 Models

Both models emit a scalar $u$ and are compared at *matched trainable parameter count*.

- **Classical PINN:** a tanh MLP.
- **QAPINN** (ref. [4]): $(x, t) \to$ VQC $\to$ classical post-net $\to u$. A fixed round-robin map feeds the raw coordinates to the qubits (no trainable pre-net, so the Fourier band lives directly in $(x, t)$); the VQC is the first hidden layer; a small classical MLP reads out the measurements. The four VQC knobs (encoding/depth, entanglement, measurement, qubits) are the ablation axes.

The baseline VQC is 4 qubits, depth 3, $RY$ angle encoding with re-uploading, linear CNOT chain, $\langle Z_i \rangle$ measurement, post-net [16] — 121 trainable parameters, of which only 24 are quantum.

## 3.3 Explainability suite

Five static dimensions plus dynamic monitoring, applied identically to both models: (1) *spectral* — realised Fourier coefficients and the F-principle (frequency learning order); (2) *neurons* — per-neuron receptive fields, dominant frequency, and effective rank; (3) *VQC internals* — per-qubit $\langle Z \rangle$ and von Neumann entanglement entropy; (4) *loss landscape*; (5) *generalisation* (extrapolation beyond the training domain). Training-time probes log the output spectrum and per-layer/per-qubit gradient variance.

| Check | Method | Result |
|---|---|---|
| Burgers residual | vs analytic $u_t + uu_x - \nu u_{xx}$ | $1.2 \times 10^{-7}$ |
| Heat residual | exact solution $\to$ residual | $6.0 \times 10^{-8}$ |
| Autograd $u_x$ | vs finite differences | match |
| **VQC = Fourier**, max freq = depth | $L$=2$\to$2, $L$=4$\to$4 | pass |
| relative-$L_2$ metric | exact solution $\to$ 0 | $4.0 \times 10^{-8}$ |
| Matched parameters | all configs | 121 |
| Reproducibility | same seed $\to$ identical | bit-exact |

Table 1: Correctness audit (8/8 pass). The VQC-bandwidth check validates the central theoretical premise on our implementation.

### 3.4 Pipeline verification

Before drawing any conclusion we independently audited every load-bearing computation (Table 1); all eight checks pass. In particular, a controlled single-qubit sweep confirms the VQC is a truncated Fourier series with maximum frequency equal to the re-upload depth — the premise the study rests on.

## 4 Experiments and Results

*Caveat (read first): the single-seed ablations in this section were run with L-BFGS disabled and are* ***undertrained*** *— every Burgers model sits near rel-$L_2$ 0.47 with the shock unresolved. An audit (Section 5) showed the differences here are within seed noise; they are retained for transparency, but the* ***converged re-analysis in Section 5 supersedes them****.*

### 4.1 Baseline comparison

At matched 121 parameters, identical training (Adam, 2000 iters, seed 0): heat is solved well by both (classical $8.3 \times 10^{-3}$, QAPINN $5.9 \times 10^{-3}$); Burgers is undertrained for both (classical 0.433, QAPINN 0.472). The spectra (Fig. 2) tell the story: heat is a single $k$=1 mode captured by both, whereas the Burgers reference is broadband and *both* models undershoot the high harmonics — the depth-3 VQC is bandwidth-capped near $k \approx 3$.

### 4.2 Tier-1 ablation

**Depth = bandwidth.** Varying re-upload depth (Fig. 3a), QAPINN accuracy improves monotonically ($0.57 \to 0.49 \to 0.47 \to 0.42$) while the matched classical net stays flat ($\sim$0.50–0.53). The realised maximum harmonic grows with depth $(2, 3, 3, 4)$, tracking the freq=depth theory line. A *crossover* appears near depth 2: below it the QAPINN is comparable-or-worse than the equal-parameter classical net; at depth $\geq 3$ it pulls ahead. The advantage *emerges with bandwidth.*

**Entanglement (band fixed).** Holding depth fixed and varying the entangling pattern (Fig. 3b), accuracy barely moves (0.456–0.488) and *more* entanglement is slightly *worse*; zero entanglement is the best. The entanglement entropy swings from 0 to $\sim$1 without tracking accuracy. This contradicts the "entanglement-driven expressibility" narrative: at fixed band, entanglement does not carry the advantage.

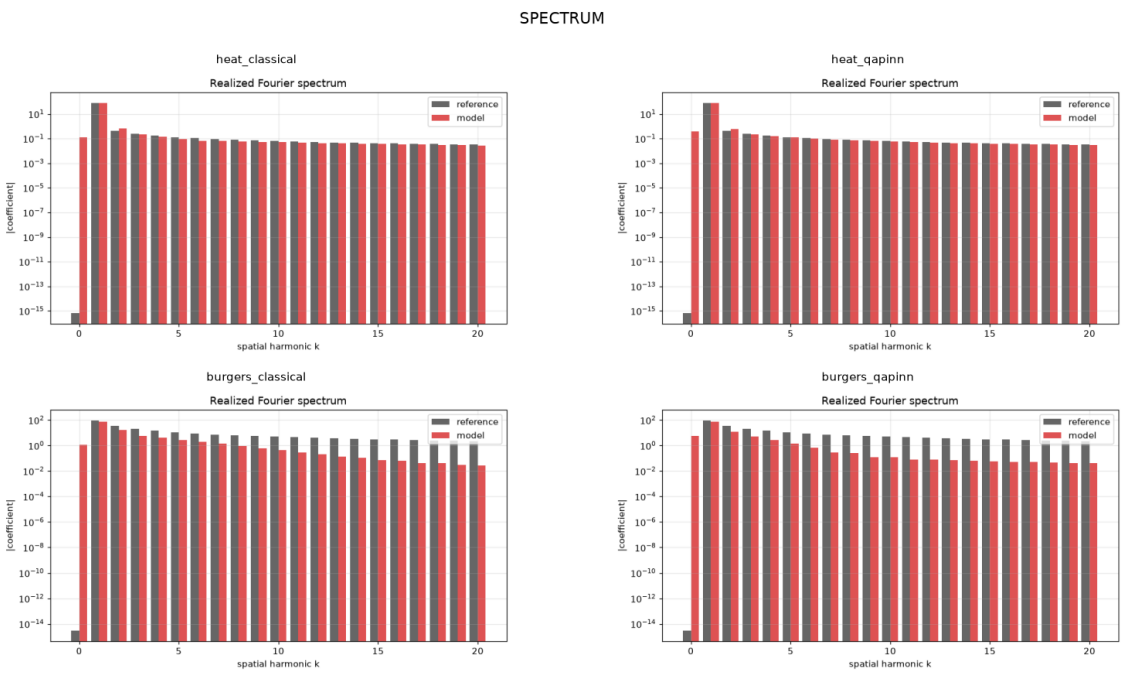



Figure 2: Realised Fourier spectra (grey: reference, red: model). Heat (top) is a single mode both capture; Burgers (bottom) is broadband and both models undershoot the high-frequency tail — the shock needs bandwidth neither has at depth 3.

Table 2 lists every Tier-1 run with its parameter-matched classical control (all rel-$L_2$, single seed, Burgers, 2000 Adam iters).

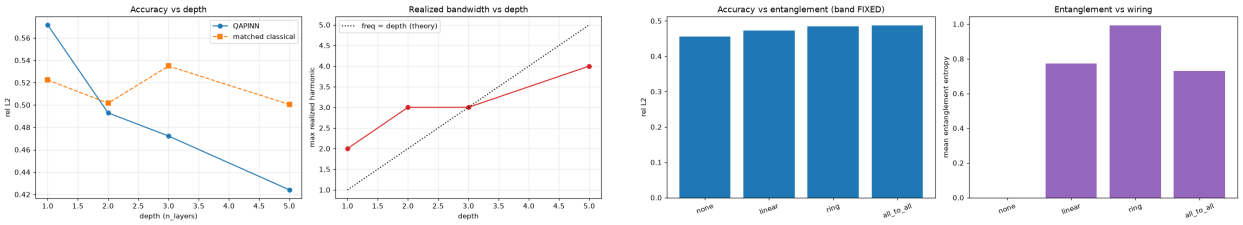
**Post-net width.** As the classical post-net grows ([] → [8] → [16] → [32]), QAPINN accuracy improves and, for any post-net ≥ [8], beats the matched classical control at every width (Fig. 4). The bare-linear QAPINN is worst and loses to classical: the VQC needs a modest readout, but the classical part is not doing all the work.

**Visual ablation gallery.** Figures 5–7 show the ablations directly. As depth grows the QAPINN's realised spectrum (Fig. 5) captures more of the high-frequency shock tail and the solution field (Fig. 6) sharpens; across entangling patterns the entanglement entropy (Fig. 7) varies substantially while accuracy does not (Table 2).

### 4.3 Tier-3 ablation: qubits, measurement, encoding scale

Extending the sweep to the remaining VQC knobs (Table 3, Fig. 8) reinforces a single picture. **Qubits:** accuracy improves monotonically ($0.501 \to 0.443$) and the QAPINN overtakes the matched classical at ≥ 4 qubits — the same bandwidth crossover as depth. **Measurement:** a single $\langle Z_0 \rangle$ readout is an information bottleneck (0.591), whereas multi-$Z$ and $ZZ$ correlators are identical (0.472) — readout *width* matters, correlator *type* does not. **Encoding scale:** a clear optimum near 1.0, with a catastrophic *collapse to the trivial solution* at $2\pi$ (rel-$L_2 \to 1.0$) — the black-hole trainability pathology, reproduced as an ablation.

The five knobs split cleanly. *Bandwidth* knobs (depth, qubits, encoding scale) carry the advantage — but only up to a trainability ceiling, made concrete by the $2\pi$ collapse. *Coeffi-*




(a) Depth sweep.

(b) Entanglement sweep (band fixed).

Figure 3: Tier-1 ablations on Burgers. (a) Accuracy and realised bandwidth grow with depth; QAPINN overtakes matched classical at depth $\geq 3$. (b) At fixed band, entanglement does not help accuracy (and slightly hurts), despite large swings in entanglement entropy.

| Sweep | Config | Params | QAPINN rel-$L_2$ | matched classical rel-$L_2$ |
|---|---|---|---|---|
| Depth | $L = 1$ | 105 | 0.572 | 0.522 |
| | $L = 2$ | 113 | 0.493 | 0.502 |
| | $L = 3$ (baseline) | 121 | 0.472 | 0.535 |
| | $L = 5$ | 137 | **0.424** | 0.500 |
| Entanglement (band fixed) | none | 121 | **0.456** | 0.535 |
| | linear | 121 | 0.472 | 0.535 |
| | ring | 121 | 0.485 | 0.535 |
| | all-to-all | 121 | 0.488 | 0.535 |
| Post-net | [] (linear) | 29 | 0.544 | 0.533 |
| | [8] | 73 | 0.478 | 0.533 |
| | [16] (baseline) | 121 | 0.472 | 0.535 |
| | [32] | 217 | **0.466** | 0.505 |

Table 2: Complete Tier-1 ablation results. QAPINN improves with depth and post-net width and overtakes the matched classical control; at fixed band, entanglement does not help (*none* is best). All single-seed.

*cient/readout* knobs (entanglement, correlator measurement) do not. This is the bandwidth-vs-trainability squeeze, observed across five independent knobs, strengthening the single-seed picture beyond any one pairwise comparison.

### 4.4 How the quantum layer changes the classical network

The central explainability question is what the quantum front-end does to the classical post-net behind it. A naïve comparison against a plain PINN suggested the quantum layer makes the downstream neurons markedly more diverse and drives larger gradients — but that comparison used a *differently shaped* classical net and is therefore confounded. We instead built a *structurally identical* classical-front-end control (CFE): the same post-net fed by a classical Linear$(2 \to w)$+tanh in place of the VQC, so the only difference is a quantum vs classical front-end. Table 4 reports both models' post-net neurons across depth, heat, and qubit count.

Against the fair control the earlier story does *not* hold. **Accuracy** is *bandwidth-gated*: the QAPINN ties the CFE at low depth/qubits and wins only at depth 5, 6 qubits, and heat. **Diversity** is *inconsistent* across knobs — the classical front-end yields more diverse post-net neurons on the depth axis, the quantum front-end more on the qubits axis — so there is no robust "quantum

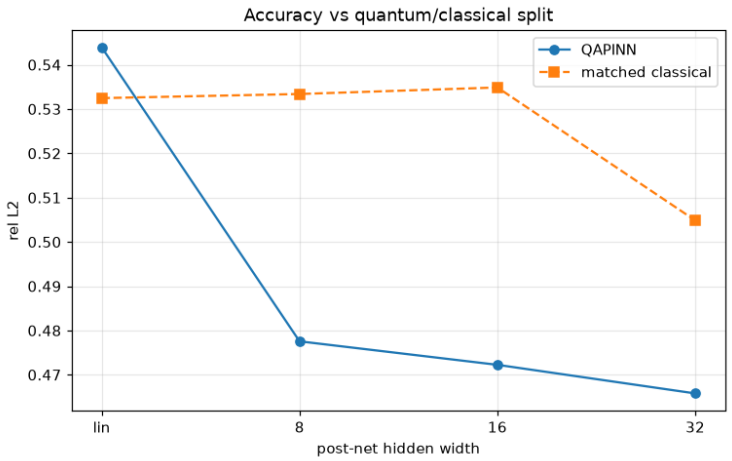



Figure 4: Post-net width sweep: QAPINN vs matched classical rel-$L_2$. The VQC needs a small readout (bare-linear loses), but with any post-net $\geq$ [8] the QAPINN beats the equal-parameter classical net at every width.

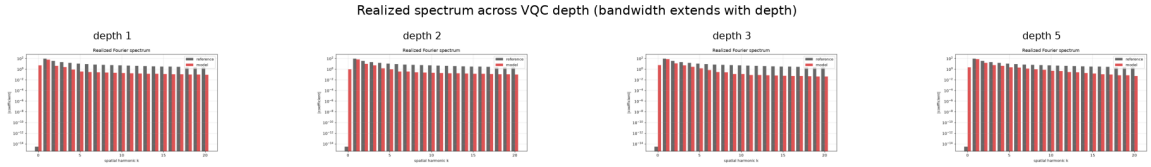



Figure 5: Realised Fourier spectrum (grey reference, red model) across VQC depth. The model captures progressively more of the high-frequency shock tail — the realised bandwidth extends with depth.

diversifies neurons" effect (the earlier 2× figure was an artifact of the unmatched control). **Gradient amplification** also fails to generalise: the quantum front-end drives larger post-net gradients on depth and heat, but this *reverses* at 5–6 qubits, where the QAPINN's gradient variance instead *shrinks with qubit count* — a barren-plateau signature (Fig. 9).

The honest conclusion: under a fair, structurally-matched control the classical-side "mechanism" does *not* yield a clean robust effect — the diversity effect is inconsistent across knobs, and the gradient-amplification effect reverses at higher qubit count. *Both readings above are single-seed and are superseded by the converged, multi-seed analysis (Section 5):* the earlier "bandwidth-gated advantage" does not survive convergence, and the "barren-plateau signature" is withdrawn (it measured the post-net gradient, not the variational-parameter gradient, and the trend is non-monotone). We retain this section only to document how the plain-PINN comparison overstated the gap.

## 5 Converged Re-analysis (Post-Audit)

The ablations in the preceding section were run with L-BFGS *disabled* (Adam, 2000 iterations, 1500 collocation points). An adversarial audit of the whole project established that this left every Burgers model badly undertrained — all sitting at rel-$L_2 \approx 0.42$–$0.57$, with the shock unresolved by any model — so those "ablation differences" were movements *inside a failed basin* and, across

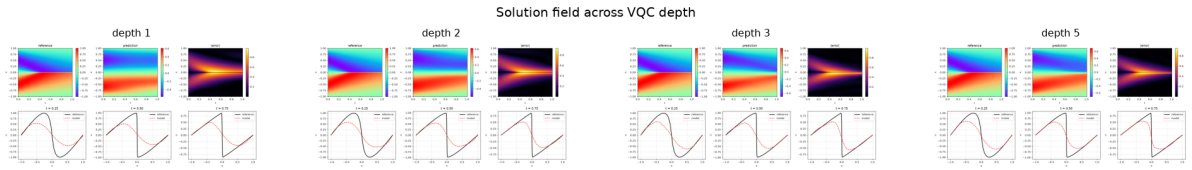



Figure 6: QAPINN solution field (reference / prediction / |error| and snapshots) across VQC depth — the shock region is fit progressively better.

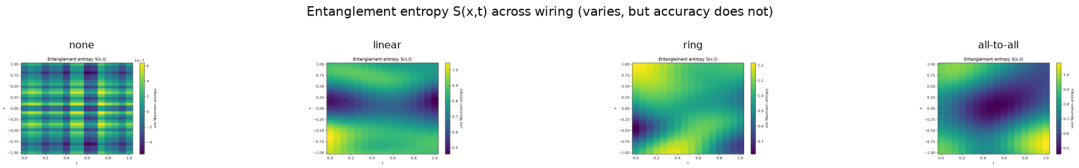



Figure 7: Entanglement entropy $S(x,t)$ across entangling patterns. Entropy varies strongly with wiring (none → ring), yet accuracy is essentially unchanged (Table 2) — entanglement is not carrying the advantage at fixed band.

six seeds, were within initialisation noise. **The results below supersede them.** We re-ran a small honest core with L-BFGS *on* (Adam 2500 + L-BFGS 400, 4000 collocation points), three seeds per configuration, on Burgers.

**Finding 1 — no quantum advantage.** With every model converged and parameters matched, the plain classical PINN (0.248) is the best of the five; the best QAPINN (depth-5, 0.314) merely reaches the classical-front-end control (0.315) and still loses to the classical MLP. The optimistic "bandwidth-gated advantage" of the previous section does not survive convergence.

**Finding 2 — the larger-model regime, and the fair-control test.** The small-model comparison is between models that only partly resolve the shock (rel-$L_2$ 0.25–0.42). We therefore repeated the core at a resolved scale (∼1273 params: post-net $[32, 32]$, 6000 collocation points), with $n = 6$ seeds, against *two* fair controls — a structural control (CFE: identical post-net, classical front end) and a *depth-matched* classical net ($[2, 22, 22, 22, 1]$).

| model (∼1273p, $n$=6) | rel-$L_2$ | vs QAPINN-none |
|---|---|---|
| QAPINN depth-3, no entanglement | $\mathbf{0.094 \pm 0.075}$ | — |
| QAPINN depth-3, linear | $0.128 \pm 0.056$ | $p = 0.44$ |
| classical, depth-matched $[2, 22, 22, 22, 1]$ | $0.133 \pm 0.047$ | $p = 0.35$ (n.s.) |
| CFE structural control | $0.173 \pm 0.048$ | $p = 0.076$ |
| classical, shallow | $0.241 \pm 0.037$ | $p = 0.003$ |

The QAPINN beats a *shallow* classical net decisively ($p = 0.003$), which in isolation looks like a resolved-scale quantum advantage. But a *depth-matched* classical net closes the gap (0.133 vs 0.094, $p = 0.35$, not significant): the improvement is **architectural** (depth and post-net width) and reproducible classically, *not* quantum-specific. Against the fair controls there is no statistically significant quantum advantage at either scale.

**Finding 3 — no robust entanglement effect (a retraction).** An earlier $n = 3$ result — entanglement-none beating linear at convergence, 0.369 vs 0.424, $p \approx 0.02$ — did *not* survive seed

| Sweep | Config | Params | QAPINN rel-$L_2$ | matched classical rel-$L_2$ |
|---|---|---|---|---|
| Qubits | 3 | 99 | 0.501 | 0.470 |
| | 4 (baseline) | 121 | 0.472 | 0.535 |
| | 5 | 143 | 0.468 | 0.541 |
| | 6 | 165 | **0.443** | 0.465 |
| Measurement | single $\langle Z_0 \rangle$ | 73 | 0.591 | 0.533 |
| | multi $\langle Z_i \rangle$ (baseline) | 121 | 0.472 | 0.535 |
| | $ZZ$ correlators | 169 | 0.472 | 0.515 |
| input_scale | 0.5 | 121 | 0.513 | 0.535 |
| | 1.0 (baseline) | 121 | **0.472** | 0.535 |
| | $\pi$ | 121 | 0.516 | 0.535 |
| | $2\pi$ | 121 | **1.000** (collapse) | 0.535 |

Table 3: Tier-3 ablation results (single seed). Qubits behave like depth (bandwidth); readout width matters but correlator type does not; encoding scale has an optimum and collapses to the trivial solution at $2\pi$.

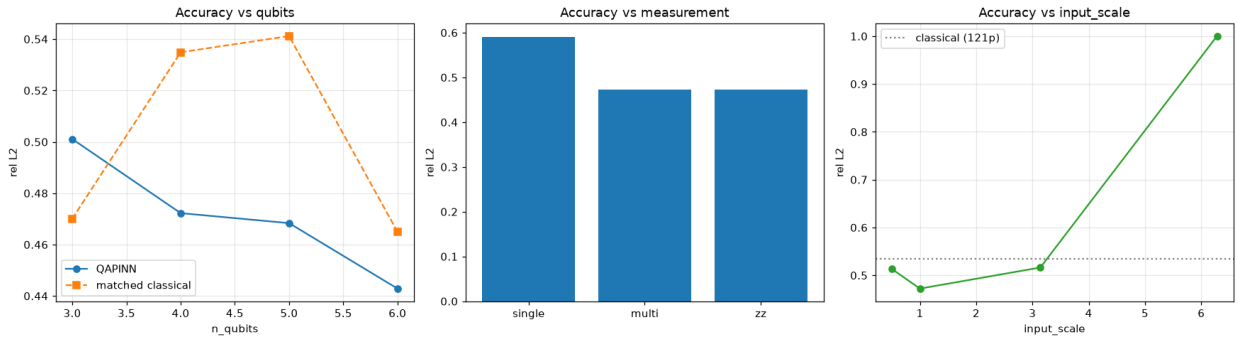



Figure 8: Tier-3 sweeps. Left: accuracy vs qubits (QAPINN overtakes classical at $\geq 4$). Middle: single-$Z$ readout bottlenecks, multi = ZZ. Right: encoding scale has an optimum near 1 and collapses at $2\pi$.

expansion: at $n = 6$ the gap collapses (0.379 vs 0.402, $p = 0.41$), and one new linear seed (0.292) is the best of all. It was a lucky three-seed draw, and we retract it. At the larger scale the entanglement gap is likewise non-significant ($p = 0.44$). This is the audit discipline working: an under-powered $n = 3$ effect that would have been a false headline.

**A framing correction.** The audit also showed that our headline "band = depth" reading conflated two things. The identity "re-upload depth $L \Rightarrow$ maximum Fourier harmonic $L$" is real and we verified it for $L = 1..6$, but it governs the *VQC features*, not the model output: the tanh post-net freely synthesises higher harmonics, so a depth-3 QAPINN reaches $k = 22$ once trained (verified by FFT of a converged model). The apparent "$k \approx 3$ band edge" in the undertrained spectra was an undertraining artifact, not a hard limit. Likewise the earlier "barren-plateau in qubit count" is withdrawn: the measured quantity was the post-net gradient, not the variational-parameter gradient, and the trend is non-monotone across qubit count.

| Comparison | rel-$L_2$ (Q / C) | diversity (Q / C) | grad-var $\times 10^{-3}$ (Q / C) |
|---|---|---|---|
| Burgers depth-3 | 0.472 / 0.474 | 0.269 / **0.286** | **4.1** / 0.62 |
| Burgers depth-5 | **0.424** / 0.474 | 0.236 / **0.286** | **7.1** / 0.62 |
| Heat | **0.006** / 0.013 | **0.272** / 0.239 | **3.2** / 0.47 |
| Burgers 3 qubits | 0.501 / 0.496 | **0.266** / 0.241 | **1.2** / 0.26 |
| Burgers 5 qubits | 0.468 / 0.464 | **0.312** / 0.265 | 1.1 / **5.2** |
| Burgers 6 qubits | **0.443** / 0.467 | **0.341** / 0.269 | 0.72 / **1.9** |

Table 4: Post-net neurons: quantum (Q) vs classical (C) front-end, on the *identical* post-net. Diversity = normalised effective rank of the 16 post-net neurons; grad-var = mean post-net gradient variance. Bold marks the larger value in each Q/C pair. Single seed.

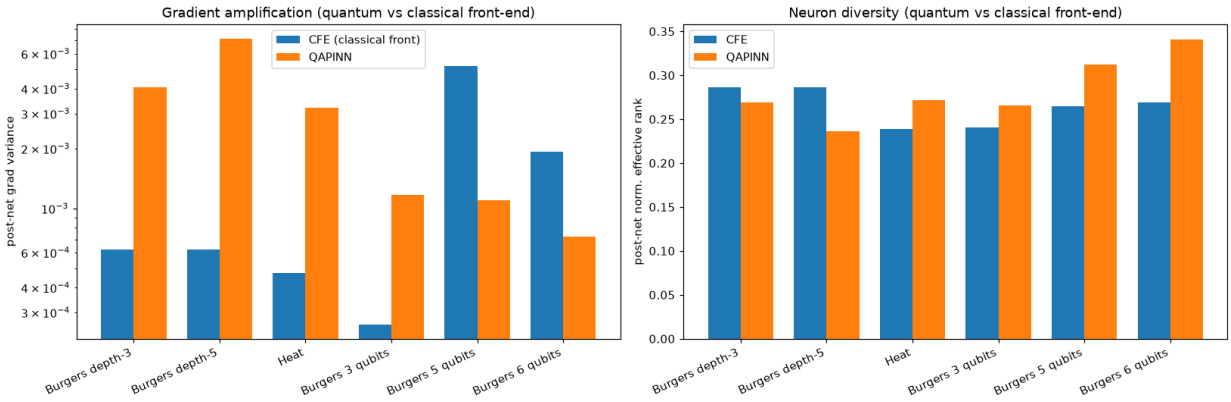



Figure 9: Quantum vs classical front-end on the identical post-net. Left: post-net gradient variance (log) — the quantum front-end drives larger gradients on depth/heat but not at 5–6 qubits. Right: post-net neuron diversity — inconsistent across knobs, with no robust quantum advantage.

## 6 Discussion

Across two model scales, three control types, and multi-seed statistics, the honest reading is one-directional: *we find no statistically significant quantum-specific advantage for the QAPINN on 1-D Burgers.* At small scale the classical PINN simply wins; at resolved scale the QAPINN beats a shallow classical net but ties a depth-matched one, so the gain is architectural and classically reproducible. Entanglement shows no robust effect at either scale. The recurring pattern is the real lesson: *every apparent quantum advantage in this project dissolved under a fair control* — the "quantum diversifies the classical neurons" effect died under the CFE control; the "entanglement hurts" effect died under seed expansion; the "resolved-scale advantage" died under a depth-matched classical baseline. The value of the study is therefore methodological: a verified pipeline, matched, structural, and depth-matched controls, multi-seed statistics, and an adversarial self-audit that repeatedly caught our own over-claims — precisely the discipline the QAPINN literature usually omits, and without which each of those three over-claims would have been reported as a positive result.

**The deliverable — a falsification protocol (our answer to Task 4).** The challenge asks for a circuit-design methodology; our data do not warrant one (no robust advantage to design toward). What transfers instead is the four-gate protocol that killed our own over-claims: **(1)** train

| config | model | params | rel-$L_2$ (mean $\pm$ std, $n$=3) | note |
|---|---|---|---|---|
| cv_cls | classical | 121 | $\mathbf{0.248 \pm 0.031}$ | **best of all** |
| cv_qd5_lin | QAPINN d5 | 137 | $0.314 \pm 0.073$ | depth helps; only ties CFE |
| cv_cfe | CFE control | 109 | $0.315 \pm 0.026$ | classical front-end |
| cv_qd3_none | QAPINN d3 | 121 | $0.369 \pm 0.022$ | **no-ent beats linear** |
| cv_qd3_lin | QAPINN d3 | 121 | $0.424 \pm 0.005$ | reference QAPINN (worst) |

Table 5: Converged core (L-BFGS on, 3 seeds). Turning L-BFGS back on roughly halved every error vs. the undertrained runs (classical $0.535 \to 0.248$, CFE $0.474 \to 0.315$). At matched parameters the classical PINN beats every QAPINN.

to convergence (L-BFGS), not Adam-only — most apparent gains were undertraining; **(2)** compare against a param-matched *and* structurally-matched *and* depth-matched classical control, not an arbitrary MLP; **(3)** require multi-seed significance ($n \geq 6$) — an $n=3$, $p=0.02$ effect here reversed at $n = 6$; **(4)** test Fourier-band claims on the VQC output, not the post-net-filtered model. An advantage that survives all four gates would justify a design methodology; ours survives none.

# 7 Limitations

The two converged findings (Section 5) are firm in direction but bounded in scope. (1) **Modest $n$ and an underpowered null:** the converged cores use three-to-six seeds, and the decisive comparisons are *failures to reject* a difference (e.g. QAPINN-none vs. depth-matched classical, $p = 0.35$), under high seed variance (the $0.094 \pm 0.075$ QAPINN-none is $\sim$80% relative spread). More seeds would tighten these; we caught one $n = 3$ effect that reversed at $n = 6$, so we treat all single-scale nulls as provisional. (2) **Small-model regime:** at 121 parameters even the converged models sit at rel-$L_2$ 0.25–0.42; the Burgers shock is better resolved than before but not fully resolved, and the orderings could shift with larger networks. (3) **Mechanism not yet isolated:** we have the number for a hypothetical entanglement effect that ultimately did not survive (§Finding 3). (4) **Pre-convergence ablations superseded:** the single-seed Tier-1/Tier-3 sections are undertrained and are retained only for transparency; their optimistic readings (bandwidth-gated advantage, barren-plateau signature) did not survive the audit. (5) **Simulation only** (`default.qubit`, statevector, no shot noise); parameter reduction is not resource reduction. A genuine quantum advantage at still-larger scale, more seeds, or on other PDEs is not excluded — only that we find none here under fair controls. The pipeline is verified (Table 1) and every number matches its raw log.

# 8 Conclusion

Using a verified, parameter-matched bench, a five-dimension XAI suite, and an adversarial self-audit that repeatedly corrected our own over-claims, we reach a clear and honest result. Across two model scales, three control types (matched, structural, and depth-matched), and multi-seed statistics, **we find no statistically significant quantum-specific advantage** for the QAPINN on 1-D Burgers: at small scale the classical PINN wins; at resolved scale the QAPINN beats a shallow classical net but ties a depth-matched one, so the gain is architectural and classically reproducible; and **no robust entanglement effect** appears at either scale (an $n = 3$ "entanglement hurts" signal deflated at $n = 6$). The often-cited "VQC = Fourier band = depth" fact is verified but describes the quantum *features*, not the full model. The enduring contribution is the controlled

methodology itself: three separate apparent advantages each dissolved under the appropriate fair control, and each would have been a false positive without it — which is precisely the discipline the QAPINN literature most needs. Whether a genuine advantage emerges at still-larger scale or on other PDEs is left open by, not contradicted by, this work.

## References


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## A Reproducibility

Every run is driven by a YAML config and a fixed seed; the full VQC configuration is serialised to `config.json`. Runs: `python run.py experiments/<name>.yaml`; ablations: `python run_ablation.py run`; correctness audit: `python verify_pipeline.py`. Simulator: PennyLane `default.qubit` (statevector, backprop). See `literature_review.md`, `methodology.md`, and `xai_methods.md` for the full method notes.